

# Inequalities – 2

## Hölder, Minkowski and the power means

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### Theoretical minimum.

**Theorem 1 (Young, Hölder, Minkowski).** For  $p, q > 1$  with  $\frac{1}{p} + \frac{1}{q} = 1$ : Young's inequality  $ab \leq \frac{a^p}{p} + \frac{b^q}{q}$  gives Hölder's inequality  $\sum |a_i b_i| \leq (\sum |a_i|^p)^{1/p} (\sum |b_i|^q)^{1/q}$ , and Hölder gives Minkowski's inequality  $\|f + g\|_p \leq \|f\|_p + \|g\|_p$ . The integral forms are identical with sums replaced by integrals;  $p = q = 2$  is Cauchy–Bunyakovsky.

**Theorem 2 (power means).** For positive  $x_i$  and weights summing to one put  $M_p = (\sum w_i x_i^p)^{1/p}$ . Then  $p \mapsto M_p$  is non-decreasing, with  $M_0 = \prod x_i^{w_i}$ ,  $M_{-\infty} = \min x_i$  and  $M_{+\infty} = \max x_i$ . The chain  $\text{HM} \leq \text{GM} \leq \text{AM} \leq \text{QM}$  is the special case  $p = -1, 0, 1, 2$ .

**Fact 1 (which exponent to take).** Hölder with a well-chosen  $p$  is the standard way to handle a sum of fractions with a root in the denominator: one writes  $\sum a_i = \sum a_i b_i^{-\theta} \cdot b_i^\theta$  and picks  $\theta$  so that the second factor becomes the constraint. The equality case again tells which exponent is right.

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1. Prove Young's inequality by convexity, deduce Hölder and then Minkowski, in the discrete and in the integral form, and describe the equality cases in all three.
2. Prove the power mean inequality together with the three limiting cases  $p \rightarrow 0$  and  $p \rightarrow \pm\infty$ , and deduce the weighted arithmetic–geometric mean inequality.
3. (IMO 2001) Prove that  $\frac{a}{\sqrt{a^2+8bc}} + \frac{b}{\sqrt{b^2+8ca}} + \frac{c}{\sqrt{c^2+8ab}} \geq 1$  for all positive  $a, b, c$ . This is the canonical application of Hölder with  $p = 3$ .

4. (Putnam 2003) Let  $a_i, b_i \geq 0$ . Show that  $(a_1 \cdots a_n)^{1/n} + (b_1 \cdots b_n)^{1/n} \leq ((a_1 + b_1) \cdots (a_n + b_n))^{1/n}$ , that is, the geometric mean is superadditive.
5. (Proposed problems) Let  $n \geq 3$ . Find the largest real constant  $C$  such that  $\frac{a_1 + \cdots + a_n}{\sqrt[n]{a_1 \cdots a_n}} - n \geq C \cdot \frac{(\sqrt[n]{a_1} - \sqrt[n]{a_n})^2}{\sqrt[n]{a_1 a_n}}$  for all positive  $a_1 \geq a_2 \geq \cdots \geq a_n$ .